

Charulkumar Chodvadiya

+917600005609 — charulpatel2499@gmail.com — /in/charulp2499 — /charulp2499 — Google Scholar — Portfolio

Computer vision & AI Engineer with 3.5+ years of combined industry and research experience building production AI systems across Healthcare AI, Computer Vision, Multimodal AI, and Generative AI—from dataset curation and model training to edge and cloud deployment. Published researcher with 5+ peer-reviewed publications, including ISBI 2024 and Best Practical Paper Award at AIxIA 2023, with collaborative research work at **IIT Bombay** and **IIT Hyderabad**. Specializes in image processing, fine-tuning LLMs and VLMs, Speech AI, and deploying scalable AI solutions using FastAPI, AWS, and Docker.

Skills

Programming:	Python (Primary), C++, SQL
AI/ML:	Machine Learning, Deep Learning, Generative AI, Multimodal AI, Document Intelligence, Speech AI, Time Series & Statistical Analysis
Foundation Models & Gen AI:	Large Language Models (LLMs), Vision-Language Models (VLMs), Transformers, Vision Transformers (ViTs), Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Prompt Engineering, RLHF, RLAI, PEFT (LoRA, QLoRA), Diffusion Models
Computer Vision:	Medical & Multi-Spectral Imaging, Image Processing, Object Detection, Image Segmentation, Contrastive Learning, Federated Learning, Transfer Learning, Anomaly Detection, GANs
Frameworks & Libraries:	PyTorch, TensorFlow, Hugging Face, OpenCV, Detectron2, MMDetection, LangChain, LangGraph, NumPy, Pandas, Scikit-learn
MLOps & Deployment:	FastAPI, Docker, AWS (EC2, Lambda, Bedrock), Kubernetes, MLflow, vLLM, Weights & Biases (W&B), TensorRT, ONNX Runtime, CUDA Optimization, Git, Linux, CI/CD
Databases:	PostgreSQL, MongoDB, Pinecone, Weaviate

Experience

AI Engineer

Sep 2025 – Now

Radin Health

- Designed and deployed an asynchronous intelligent multi-agent AI copilot automating healthcare front-desk operations, including patient onboarding, document verification, order creation, scheduling calls, pending action resolution, and follow-up workflows, improving operational efficiency across high-volume clinical processes.
- Built token-optimized LLM pipelines processing 18,000+ clinical records daily for automated QA, MIPS compliance validation, CPT/RVU values & ICD-10 code extraction, and AI-generated follow-up recommendations; reduced LLM inference costs by 20–30% through prompt compression while maintaining ~94–98% extraction accuracy.
- Fine-tuned and deployed Vision-Language Models (VLMs) for medical document intelligence, achieving 96–99% metadata extraction accuracy across 4,000+ multi-page healthcare documents daily; engineered scalable asynchronous APIs with background task offloading to minimize user-perceived latency.
- Developed and deployed real-time patient-facing AI voice bots and conversational chatbots for appointment booking, rescheduling, cancellations, and patient engagement, integrating WebSocket-based STT/TTS pipelines with 10–15 ms streaming latency and achieving a 92% end-to-end task success rate.
- Fine-tuned domain-specific speech recognition models for medical dictation, improving recognition of clinical terminology, anatomical entities, dates, and quantitative measurements, reducing transcription and reporting errors by ~80% compared to baseline speech recognition systems.

AI Developer

Jan 2025 – Aug 2025

HubBroker ApS

- Fine-tuned and deployed LoRA-based LLMs and VLMs for multilingual European business document intelligence, achieving 94% extraction accuracy on structured, semi-structured, and tabular documents.
- Developed AI-driven document quality classification systems achieving 99% accuracy across diverse business document types, enabling automated validation and processing workflows at scale.
- Designed and deployed production-grade FastAPI services integrating LLMs and VLMs into enterprise automation pipelines, incorporating GPU-optimized inference and scalable deployment workflows.

AI Research Intern

Jun 2023 – Apr 2024

Indian Institute of Technology Hyderabad (VIGIL lab)

- Researched brain tumor segmentation using federated learning across decentralized medical imaging datasets.
- Proposed and implemented a contrastive learning-based local training loss to mitigate data heterogeneity across multiple clients, achieving a 5–12% improvement in segmentation accuracy.

Research Collaboration

AI Researcher

Koita Center of Digital Health (IITB)

- Participated in diverse medical AI projects, including medical imaging and EEG signal analysis. Authored publications and contributed to peer-review processes for other fellow researchers' work.
- Developed a novel loss function for enhancing medical image segmentation accuracy by 6-8%, leading to a publication at A-tier conference - ISBI24.

Education

Masters in Technology - Artificial Intelligence

(Gold Medalist) (CGPA: 8.94 / 10.00)

Pandit Deendayal Energy University

Aug 2022 – May 2024

- Relevant Coursework: Computer Vision, Deep Neural Networks, Machine Learning, Time Series Data Analysis, Pattern Recognition and Machine Learning, Explainable AI, NLP, Cloud Infrastructure for AI
- **Teaching Assistant (Research):** Conducted research in machine learning, statistical data modeling, and computer vision. Key projects included solar power forecasting and maintenance prediction, night vision face detection, and palm print identification.
- IEEE CIS Student Branch Chapter Chair / IEEE CS Student Branch Chapter Vice-chair

Bachelor of Engineering - Computer Engineering

(First Class with Distinction) (CPI: 7.95 / 10.00)

Gujarat Technological University

Jul 2017 – Jun 2021

Selected Projects

Medical Image Segmentation — Centralized & Federated

[Code \(Centralized\)](#) | [Code \(Federated\)](#)

- Developed 3D brain tumor segmentation pipelines on multi-institutional MRI datasets in both centralized and federated learning settings; proposed a novel Feature-Enhanced Spatial Segmentation (FeSS) loss function that improved segmentation accuracy by 6–8%, resulting in a publication at IEEE ISBI 2024.
- Implemented federated learning across decentralized medical imaging clients using a contrastive learning-based local training objective, mitigating data heterogeneity and improving segmentation performance by 5–12%.

SynthAI: Interactive Generative Image Sandbox

[Live Space](#)

- Built an interactive web platform on Hugging Face Spaces enabling real-time image generation and side-by-side comparison across multiple GAN architectures, designed for research exploration and generative model benchmarking.

Real-Time Depth Estimation & 3D Point Cloud Reconstruction

[Code](#) | [Demo](#)

- Developed a real-time depth estimation system combining MiDaS depth prediction and YOLOv5 object detection on live webcam streams, enabling 3D scene understanding and point cloud reconstruction for spatial perception applications..

PalmScanX: Contactless Palmprint Biometric Authentication

[Code](#)

- Created a proprietary palmprint dataset comprising 3,600+ annotated images and developed contactless biometric authentication pipelines using YOLOv5 and YOLOv8, incorporating image enhancement, noise reduction, and transfer learning techniques for robust palmprint detection.

Publications

- [C. Chodvadiya](#), N Mahla, KG Singh, and KS Jadhav, "FESS Loss: Feature-Enhanced Spatial Segmentation Loss for Optimizing Medical Image Analysis," *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*. [\[Link\]](#)
- [C. Chodvadiya](#), V. Solanki, and KG Singh, "Intelligent Virtual Worlds: A Survey of the Role of AI in the Metaverse," 2024 3rd International Conference for Innovation in Technology (INOCON). [\[Link\]](#)
- SK Gaurav, [C. Chodvadiya](#), B Chintan, S Pooja, and A Bruno, "Seeing in the Dark: A Different Approach to Night Vision Face Detection with Thermal IR Images," (2023) CEUR-WS under AIXIA. [\[Best Practical Paper\]](#) [\[Link\]](#)
- [C. Chodvadiya](#), KG Singh, and R Vyas, "Detection of palmprint from contactless smartphone-based video hand dataset." NCVPRIPG 2023. Singapore: Springer Nature Singapore, 2023. [\[Link\]](#)
- KG Singh, [C. Chodvadiya](#), S Vyas, H Patel, S Mishra, and B Tripathi, "Prediction of Solar Power Generation and Maintenance Activities for 1MW Power Plant," 2023 IEEE IAS Global Conference on Emerging Technologies (GlobConET), 2023. [\[Link\]](#)